

RISHI SHAH

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Education

McMaster University

B.Eng. Computer Engineering | Schulich Leader Scholar (\$120K, 1 of 50 in Canada)

Apr. 2029

Hamilton, Ontario

Experience

E3A Healthcare

May 2026 – Aug 2026

Machine Learning Engineer

Shenzhen, China

- Owned architecture direction for the flagship labor-prediction model, turning at-home fetal monitoring into a posterior over days to delivery; it dates term delivery within 3 days, 3x tighter than the 11-day clinical SOTA
- Designed the architecture the research team now builds on: cohort mixture-of-experts with a learned gate, quantile heads for calibrated uncertainty, a reasoning layer, and the sub-agent decomposition now on the roadmap
- Built the supervised layer and coupled-likelihood objective behind the preterm win, halving error on the highest-risk births by 2.02 days
- Authored the evaluation methodology the whole project now reports against and automated multi-region GPU cluster provisioning on GCP, putting on-demand multi-node training in front of every engineer on the team

The Centre for Mechatronics and Hybrid Technologies

Oct. 2025 – Apr. 2026

Research Engineer

Hamilton, ON

- Cut battery voltage-model test RMSE 84% (12.82 mV to 2.03 mV) with a physics-ML hybrid (PyBaMM SPMe + seq2seq LSTM residual), then shipped it through ONNX to a real-time Simulink BMS block at bit-exact parity
- Implemented Extended Kalman Filter and SVSF state-of-charge estimators from scratch in NumPy with a genetic-algorithm 2RC-ECM parameter ID (0.58 mV mean HPPC RMSE); co-authored paper under submission

Western University

Apr. 2024 – Aug. 2025

Machine Learning Researcher

London, ON

- Architected a recurrent, denoising CORnet-S variant that beats MIT's CORnet-S on robustness with no adversarial training: 97.82% PGD on MNIST, 89.3% on CIFAR-100, 84.7% on ImageNet100 at ~15% extra compute

Robarts Research Institute

Nov. 2022 – May 2024

Computer Vision Researcher

London, ON

- Trained a U-Net with Synaptive Medical for surgical-tool segmentation on live OR video: 95% DICE, <0.12 s/frame

Projects & Preprints

Lethe | *CUDA, Triton, PyTorch* | *first-author preprint*

2026

- Wrote the first native Blackwell tcgen05 training backward for the gated-linear-recurrence family (Mamba-2, GLA, KDA, Gated DeltaNet), plus the first open Triton Mamba-3 MIMO backward: 3.3×10^{-3} worst relative error against an fp64 oracle, bit-for-bit deterministic
- Built a 12-gate contract-grade verifier (tolerance-free gates, derived bounds, a 19-kernel red team) that audited 2,638 LLM-generated kernels a public benchmark had accepted and falsified 62.1%

RDR: Rollout-Decoded Reconstruction | *PyTorch* | *first-author preprint*

2026

- Nearly doubled the horizon a latent world model stays accurate over, for free: one loss term that free-runs the model during training, decodes every rollout latent, and penalizes drift against ground truth, at zero added parameters
- Lifted valid prediction time 1.80x on the chaotic Kuramoto-Sivashinsky equation (3.87 to 6.97 time units) at identical 193,568 parameters and identical inference cost, winning 10 of 10 preregistered configurations

CGM-on-Chip | *C++17*

May 2026

- Implemented a complete deep-learning stack in ~5,500 lines of pure C++17 (from-scratch reverse-mode autograd, hand-rolled Adam, fork-join training thread pool) and fixed a box-projection clamp missing from OSDN's algorithm
- Delivered 60-minute-ahead hypoglycemia prediction (test AUROC 0.9113) on a \$4 ESP32-S3 using 20% of its RAM, matching the FP64 trainer to 8×10^{-7}

Sentinel | *TypeScript, Node.js LSP, C++* | *Best Developer Tool at HackWestern 12*

Nov. 2025

- Solo-built a hardware-aware static analyzer with a custom LSP server running 50+ silicon-level checks (RAM/Flash budgets, strapping-pin hazards, bus conflicts) at ~47 ms latency; 1.2K+ installs on VS Code and Open VSX

Technical Skills

Languages: Python, C++17, CUDA, Triton, TypeScript, MATLAB, Java, SQL, Bash

Frameworks & Infra: PyTorch, JAX, CUTLASS, ONNX, Simulink, Node.js, GCP (B200/H100), AWS, Docker, Linux, Git